

Groups and Lie Algebras: Solutions to Exercise Sheet 8

We use the physics convention. The irreducible representation of spin j has dimension $2j + 1$ and is denoted V_j . It has a standard basis

$$\{|j, m\rangle \mid m = -j, -j + 1, \dots, j\}.$$

The generators H, E, F of $\mathfrak{sl}(2)$ satisfy

$$[H, E] = 2E, \quad [H, F] = -2F, \quad [E, F] = H,$$

and act by

$$\begin{aligned} H|j, m\rangle &= 2m|j, m\rangle, \\ E|j, m\rangle &= \sqrt{(j - m)(j + m + 1)} |j, m + 1\rangle, \\ F|j, m\rangle &= \sqrt{(j + m)(j - m + 1)} |j, m - 1\rangle. \end{aligned}$$

Thus E raises the magnetic quantum number m by one, while F lowers it by one.

Convention warning. Clebsch–Gordan coefficients are determined only up to phase choices for each irreducible summand. We choose real coefficients and fix phases by taking the highest-weight vectors indicated below with positive leading coefficient. Changing these phase choices may change some signs, but not the representation theory.

1 Part 1: Tensor product decomposition

Let V_j and $V_{j'}$ be irreducible $\mathfrak{sl}(2)$ -representations. The tensor product representation is defined by

$$X \cdot (v \otimes w) = (Xv) \otimes w + v \otimes (Xw), \quad X \in \mathfrak{sl}(2).$$

Equivalently, on the tensor product the generators are

$$\begin{aligned} H_{\text{tot}} &= H \otimes \text{id} + \text{id} \otimes H, \\ E_{\text{tot}} &= E \otimes \text{id} + \text{id} \otimes E, \quad F_{\text{tot}} = F \otimes \text{id} + \text{id} \otimes F. \end{aligned}$$

1(a). Eigenvectors of H on the tensor product

We compute the action of H on a tensor product basis vector:

$$\begin{aligned} H(|j, m\rangle \otimes |j', m'\rangle) &= (H|j, m\rangle) \otimes |j', m'\rangle + |j, m\rangle \otimes (H|j', m'\rangle) \\ &= 2m|j, m\rangle \otimes |j', m'\rangle + 2m'|j, m\rangle \otimes |j', m'\rangle \\ &= 2(m + m')|j, m\rangle \otimes |j', m'\rangle. \end{aligned}$$

Therefore the tensor product basis vector

$$|j, m\rangle \otimes |j', m'\rangle$$

is an eigenvector of H with eigenvalue

$$2(m + m').$$

It is useful to define

$$M = m + m'.$$

Then the H -eigenvalue is $2M$. In physics language, M is the total magnetic quantum number.

1(b). Checking the Clebsch–Gordan decomposition

The claimed decomposition is

$$V_j \otimes V_{j'} \cong V_{j+j'} \oplus V_{j+j'-1} \oplus \cdots \oplus V_{|j-j'|}.$$

We verify this by comparing dimensions and H -eigenvalue multiplicities.

Dimension check. Without loss of generality, assume $j \geq j'$. Then the summands on the right-hand side have spins

$$J = j + j', j + j' - 1, \dots, j - j'.$$

There are $2j' + 1$ summands. Their total dimension is

$$\begin{aligned} \sum_{r=0}^{2j'} \dim V_{j+j'-r} &= \sum_{r=0}^{2j'} (2(j + j' - r) + 1) \\ &= (2j' + 1)(2j + 2j' + 1) - 2 \sum_{r=0}^{2j'} r \\ &= (2j' + 1)(2j + 2j' + 1) - 2 \frac{(2j')(2j' + 1)}{2} \\ &= (2j' + 1)(2j + 1). \end{aligned}$$

But

$$\dim(V_j \otimes V_{j'}) = (2j + 1)(2j' + 1),$$

so the dimensions agree.

Multiplicity check for H -eigenvalues. In the tensor product basis, the eigenvalue $2M$ occurs once for every pair (m, m') satisfying

$$-j \leq m \leq j, \quad -j' \leq m' \leq j', \quad m + m' = M.$$

Thus the multiplicity is the number of such pairs.

Again assume $j \geq j'$. The possible values of M range from $-(j + j')$ to $j + j'$ in steps of one. The multiplicity pattern is symmetric under $M \mapsto -M$. For $M \geq 0$, it is

$$\text{mult}(M) = \begin{cases} 2j' + 1, & 0 \leq M \leq j - j', \\ j + j' - M + 1, & j - j' \leq M \leq j + j'. \end{cases}$$

This is the familiar “triangle” pattern of tensor product weights.

Now compare with the right-hand side. In a single irreducible representation V_J , each weight M with $-J \leq M \leq J$ occurs exactly once. Thus in the direct sum

$$V_{j+j'} \oplus V_{j+j'-1} \oplus \cdots \oplus V_{j-j'},$$

the multiplicity of M is exactly the number of spins J in the list satisfying

$$J \geq |M|.$$

This gives precisely the same multiplicity pattern as above. Therefore the claimed decomposition has the same dimension and the same H -eigenvalue multiplicities as the tensor product. This confirms

$$V_j \otimes V_{j'} \cong V_{j+j'} \oplus V_{j+j'-1} \oplus \cdots \oplus V_{|j-j'|}.$$

2 Part 2: Clebsch–Gordan coefficients

The decomposition above gives a second natural basis of $V_j \otimes V_{j'}$:

$$\{|J, M\rangle \mid J = |j - j'|, \dots, j + j', \ M = -J, \dots, J\}.$$

The Clebsch–Gordan coefficients are defined by the expansion

$$|J, M\rangle = \sum_{m, m' : m+m'=M} \langle j, m; j', m' \mid J, M\rangle |j, m\rangle \otimes |j', m'\rangle.$$

2(a). Why only terms with $m + m' = M$ appear

The vector $|J, M\rangle$ is an eigenvector of H with eigenvalue $2M$:

$$H|J, M\rangle = 2M|J, M\rangle.$$

On the other hand, the tensor product basis vector

$$|j, m\rangle \otimes |j', m'\rangle$$

has H -eigenvalue $2(m + m')$. Since vectors with different eigenvalues of H lie in different eigenspaces, an eigenvector with eigenvalue $2M$ can only be expanded using tensor product basis vectors with the same eigenvalue. Therefore only terms satisfying

$$m + m' = M$$

can occur in the expansion.

2(b). The highest-weight vector of the top summand

The largest possible value of $M = m + m'$ is obtained by taking

$$m = j, \quad m' = j'.$$

Therefore the eigenspace with maximal weight

$$M = j + j'$$

is one-dimensional and is spanned by

$$|j, j\rangle \otimes |j', j'\rangle.$$

The top summand in the Clebsch–Gordan decomposition is $V_{j+j'}$, whose highest-weight vector has

$$J_{\max} = j + j', \quad M = J_{\max}.$$

Hence

$$|J_{\max}, J_{\max}\rangle$$

must be proportional to

$$|j, j\rangle \otimes |j', j'\rangle.$$

We fix the normalization and phase by choosing

$$\boxed{|J_{\max}, J_{\max}\rangle = |j, j\rangle \otimes |j', j'\rangle.}$$

Thus the corresponding Clebsch–Gordan coefficient is 1:

$$\langle j, j; j', j' | j + j', j + j' \rangle = 1.$$

2(c). Descendants by repeated application of F

Inside a spin- J irreducible representation, the lowering operator satisfies

$$F|J, M\rangle = \sqrt{(J+M)(J-M+1)}|J, M-1\rangle.$$

Therefore, once the highest-weight vector $|J, J\rangle$ is known, all lower weight vectors in the same irreducible representation are determined by successively applying F . Explicitly,

$$|J, M\rangle = \sqrt{\frac{(J+M)!}{(J-M)!(2J)!}} F^{J-M}|J, J\rangle,$$

where the factorial expression is understood for integer $J-M$.

On the tensor product, the lowering operator is

$$F_{\text{tot}} = F \otimes \text{id} + \text{id} \otimes F.$$

Thus if

$$|J, J\rangle$$

is written in the tensor product basis, applying F_{tot} and normalizing gives $|J, J-1\rangle$, applying it again gives $|J, J-2\rangle$, and so on. In this way, all Clebsch–Gordan coefficients in that summand are determined recursively.

For example, for the top summand $J_{\max} = j + j'$, we have

$$\begin{aligned} F_{\text{tot}}|J_{\max}, J_{\max}\rangle &= F_{\text{tot}}(|j, j\rangle \otimes |j', j'\rangle) \\ &= \sqrt{2j}|j, j-1\rangle \otimes |j', j'\rangle + \sqrt{2j'}|j, j\rangle \otimes |j', j'-1\rangle. \end{aligned}$$

Since

$$F|J_{\max}, J_{\max}\rangle = \sqrt{2J_{\max}}|J_{\max}, J_{\max}-1\rangle,$$

we get

$$\boxed{|J_{\max}, J_{\max}-1\rangle = \frac{\sqrt{2j}|j, j-1\rangle \otimes |j', j'\rangle + \sqrt{2j'}|j, j\rangle \otimes |j', j'-1\rangle}{\sqrt{2(j+j')}}.}$$

Equivalently,

$$\boxed{|J_{\max}, J_{\max}-1\rangle = \frac{\sqrt{j}|j, j-1\rangle \otimes |j', j'\rangle + \sqrt{j'}|j, j\rangle \otimes |j', j'-1\rangle}{\sqrt{j+j'}}.}$$

2(d). Finding the next highest-weight vector

The next summand is

$$V_{J_{\max}-1}.$$

Its highest-weight vector has

$$J = J_{\max} - 1 = j + j' - 1, \quad M = J = j + j' - 1.$$

The weight space with $M = j + j' - 1$ is two-dimensional, spanned by

$$a := |j, j\rangle \otimes |j', j' - 1\rangle, \quad b := |j, j - 1\rangle \otimes |j', j'\rangle.$$

One vector in this space already belongs to the top summand, namely

$$|J_{\max}, J_{\max} - 1\rangle = \frac{\sqrt{j'} a + \sqrt{j} b}{\sqrt{j + j'}}.$$

Therefore the highest-weight vector of the next summand must be orthogonal to this vector. A normalized orthogonal choice is

$$|J_{\max} - 1, J_{\max} - 1\rangle = \frac{\sqrt{j} a - \sqrt{j'} b}{\sqrt{j + j'}}.$$

That is,

$$|j + j' - 1, j + j' - 1\rangle = \frac{\sqrt{j} |j, j\rangle \otimes |j', j' - 1\rangle - \sqrt{j'} |j, j - 1\rangle \otimes |j', j'\rangle}{\sqrt{j + j'}}.$$

One can also check directly that it is a highest-weight vector by applying E_{tot} : the two terms cancel.

3 Part 3: Explicit computation for $V_{1/2} \otimes V_{1/2}$

Write

$$|+\rangle := |\tfrac{1}{2}, \tfrac{1}{2}\rangle, \quad |-\rangle := |\tfrac{1}{2}, -\tfrac{1}{2}\rangle.$$

Then

$$\begin{aligned} H|+\rangle &= |+\rangle, & H|-\rangle &= -|-\rangle, \\ E|-\rangle &= |+\rangle, & E|+\rangle &= 0, \\ F|+\rangle &= |-\rangle, & F|-\rangle &= 0. \end{aligned}$$

3(a). Tensor product basis and H -eigenvalues

The tensor product $V_{1/2} \otimes V_{1/2}$ has basis

$$|+\rangle \otimes |+\rangle, \quad |+\rangle \otimes |-\rangle, \quad |-\rangle \otimes |+\rangle, \quad |-\rangle \otimes |-\rangle.$$

Their total M -values and H -eigenvalues are:

basis vector	$M = m + m'$	H -eigenvalue = $2M$
$ +\rangle \otimes +\rangle$	1	2
$ +\rangle \otimes -\rangle$	0	0
$ -\rangle \otimes +\rangle$	0	0
$ -\rangle \otimes -\rangle$	-1	-2

3(b). Clebsch–Gordan coefficients

The decomposition is

$$V_{1/2} \otimes V_{1/2} \cong V_1 \oplus V_0.$$

The spin-1 summand has three basis vectors $|1, 1\rangle$, $|1, 0\rangle$, $|1, -1\rangle$. The spin-0 summand has one basis vector $|0, 0\rangle$.

The highest-weight vector of the spin-1 summand is

$$|1, 1\rangle = |+\rangle \otimes |+\rangle.$$

Apply $F_{\text{tot}} = F \otimes \text{id} + \text{id} \otimes F$:

$$\begin{aligned} F|1, 1\rangle &= F(|+\rangle \otimes |+\rangle) \\ &= |-\rangle \otimes |+\rangle + |+\rangle \otimes |-\rangle. \end{aligned}$$

But in V_1 ,

$$F|1, 1\rangle = \sqrt{2}|1, 0\rangle.$$

Hence

$$|1, 0\rangle = \frac{1}{\sqrt{2}}(|+\rangle \otimes |-\rangle + |-\rangle \otimes |+\rangle).$$

Applying F once more gives

$$|1, -1\rangle = |-\rangle \otimes |-\rangle.$$

The remaining vector in the $M = 0$ subspace must be orthogonal to $|1, 0\rangle$. We choose

$$|0, 0\rangle = \frac{1}{\sqrt{2}}(|+\rangle \otimes |-\rangle - |-\rangle \otimes |+\rangle).$$

This is indeed a highest-weight vector because

$$E|0, 0\rangle = 0.$$

Thus the triplet V_1 is the symmetric part, and the singlet V_0 is the antisymmetric part.

3(c). Change-of-basis matrix

Use the tensor product basis ordered as

$$|++\rangle, |+-\rangle, |-+\rangle, |--\rangle,$$

where for example $|+-\rangle = |+\rangle \otimes |-\rangle$. Use the coupled basis ordered as

$$|1, 1\rangle, |1, 0\rangle, |1, -1\rangle, |0, 0\rangle.$$

Then the change-of-basis matrix C , whose rows are the coupled basis vectors written in the tensor product basis, is

$$C = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \\ 0 & 0 & 0 & 1 \\ 0 & \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} & 0 \end{pmatrix}.$$

The rows are orthonormal, so

$$CC^T = I.$$

Since the matrix is square, this is equivalent to

$$C^T C = I.$$

Thus C is an orthogonal matrix. This simply reflects the fact that we have changed from one orthonormal basis to another.

3(d). Explicit checks

First verify that $E|0, 0\rangle = 0$:

$$\begin{aligned} E|0, 0\rangle &= \frac{1}{\sqrt{2}} E(|+\rangle \otimes |-\rangle - |-\rangle \otimes |+\rangle) \\ &= \frac{1}{\sqrt{2}} (|+\rangle \otimes |+\rangle - |+\rangle \otimes |+\rangle) \\ &= 0. \end{aligned}$$

Next verify that $F|1, -1\rangle = 0$:

$$F|1, -1\rangle = F(|-\rangle \otimes |-\rangle) = 0,$$

because $F|-\rangle = 0$ in each factor.

4 Part 4: The case $V_1 \otimes V_{1/2}$

We now consider the tensor product of the spin-1 representation and the spin-1/2 representation. Write the basis of V_1 as

$$|1, 1\rangle, \quad |1, 0\rangle, \quad |1, -1\rangle,$$

and the basis of $V_{1/2}$ as

$$|+\rangle = |\tfrac{1}{2}, \tfrac{1}{2}\rangle, \quad |-\rangle = |\tfrac{1}{2}, -\tfrac{1}{2}\rangle.$$

4(a). H -eigenvalues and multiplicities

The tensor product basis and total M -values are:

basis vector	$M = m + m'$	H -eigenvalue = $2M$
$ 1, 1\rangle \otimes +\rangle$	$\frac{3}{2}$	3
$ 1, 0\rangle \otimes +\rangle$	$\frac{1}{2}$	1
$ 1, 1\rangle \otimes -\rangle$	$\frac{1}{2}$	1
$ 1, -1\rangle \otimes +\rangle$	$-\frac{1}{2}$	-1
$ 1, 0\rangle \otimes -\rangle$	$-\frac{1}{2}$	-1
$ 1, -1\rangle \otimes -\rangle$	$-\frac{3}{2}$	-3

Therefore the eigenvalue multiplicities are

$2M$	3	1	-1	-3
multiplicity	1	2	2	1

This agrees with the decomposition

$$V_1 \otimes V_{1/2} \cong V_{3/2} \oplus V_{1/2}.$$

Indeed, $V_{3/2}$ contributes weights

$$M = \frac{3}{2}, \frac{1}{2}, -\frac{1}{2}, -\frac{3}{2},$$

and $V_{1/2}$ contributes weights

$$M = \frac{1}{2}, -\frac{1}{2}.$$

4(b). Explicit coupled basis vectors

We first construct the spin-3/2 summand. The highest-weight vector is unique:

$$\boxed{|\frac{3}{2}, \frac{3}{2}\rangle = |1, 1\rangle \otimes |+\rangle.}$$

Apply F_{tot} :

$$\begin{aligned} F|\frac{3}{2}, \frac{3}{2}\rangle &= F(|1, 1\rangle \otimes |+\rangle) \\ &= \sqrt{2}|1, 0\rangle \otimes |+\rangle + |1, 1\rangle \otimes |-\rangle. \end{aligned}$$

Since in $V_{3/2}$,

$$F|\frac{3}{2}, \frac{3}{2}\rangle = \sqrt{3}|\frac{3}{2}, \frac{1}{2}\rangle,$$

we get

$$\boxed{|\frac{3}{2}, \frac{1}{2}\rangle = \sqrt{\frac{2}{3}}|1, 0\rangle \otimes |+\rangle + \frac{1}{\sqrt{3}}|1, 1\rangle \otimes |-\rangle.}$$

Applying F again gives

$$\boxed{|\frac{3}{2}, -\frac{1}{2}\rangle = \frac{1}{\sqrt{3}}|1, -1\rangle \otimes |+\rangle + \sqrt{\frac{2}{3}}|1, 0\rangle \otimes |-\rangle.}$$

Finally,

$$\boxed{|\frac{3}{2}, -\frac{3}{2}\rangle = |1, -1\rangle \otimes |-\rangle.}$$

Now we construct the spin-1/2 summand. Its highest-weight vector has $M = 1/2$, so it lies in the two-dimensional subspace spanned by

$$|1, 0\rangle \otimes |+\rangle, \quad |1, 1\rangle \otimes |-\rangle.$$

It must be orthogonal to $|\frac{3}{2}, \frac{1}{2}\rangle$. With our phase convention we choose

$$\boxed{|\frac{1}{2}, \frac{1}{2}\rangle = \frac{1}{\sqrt{3}}|1, 0\rangle \otimes |+\rangle - \sqrt{\frac{2}{3}}|1, 1\rangle \otimes |-\rangle.}$$

One checks directly that it is killed by E , so it is indeed a highest-weight vector. Lowering it gives

$$\boxed{|\frac{1}{2}, -\frac{1}{2}\rangle = \sqrt{\frac{2}{3}}|1, -1\rangle \otimes |+\rangle - \frac{1}{\sqrt{3}}|1, 0\rangle \otimes |-\rangle.}$$

These six vectors form an orthonormal basis adapted to the irreducible decomposition

$$V_1 \otimes V_{1/2} \cong V_{3/2} \oplus V_{1/2}.$$

4(c). Clebsch–Gordan coefficient table

Order the tensor product basis as

$$(1, \frac{1}{2}), \quad (0, \frac{1}{2}), \quad (1, -\frac{1}{2}), \quad (-1, \frac{1}{2}), \quad (0, -\frac{1}{2}), \quad (-1, -\frac{1}{2}).$$

Then the Clebsch–Gordan coefficient table is:

	$(1, \frac{1}{2})$	$(0, \frac{1}{2})$	$(1, -\frac{1}{2})$	$(-1, \frac{1}{2})$	$(0, -\frac{1}{2})$	$(-1, -\frac{1}{2})$
$ \frac{3}{2}, \frac{3}{2}\rangle$	1	0	0	0	0	0
$ \frac{3}{2}, \frac{1}{2}\rangle$	0	$\sqrt{\frac{2}{3}}$	$\frac{1}{\sqrt{3}}$	0	0	0
$ \frac{3}{2}, -\frac{1}{2}\rangle$	0	0	0	$\frac{1}{\sqrt{3}}$	$\sqrt{\frac{2}{3}}$	0
$ \frac{3}{2}, -\frac{3}{2}\rangle$	0	0	0	0	0	1
$ \frac{1}{2}, \frac{1}{2}\rangle$	0	$\frac{1}{\sqrt{3}}$	$-\sqrt{\frac{2}{3}}$	0	0	0
$ \frac{1}{2}, -\frac{1}{2}\rangle$	0	0	0	$\sqrt{\frac{2}{3}}$	$-\frac{1}{\sqrt{3}}$	0

All entries not shown are zero because $M = m + m'$ must be conserved.

5 Part 5: Application to spin–orbit coupling

We now consider a particle with orbital angular momentum $\ell = 1$ and spin $s = \frac{1}{2}$. The state space is

$$V_1 \otimes V_{1/2}.$$

The orbital angular momentum acts on the first factor, and the spin angular momentum acts on the second factor.

The generators are related to angular momentum operators by

$$L_z = \frac{1}{2}H, \quad L_+ = E, \quad L_- = F.$$

The Casimir operator is

$$J^2 = L_z^2 + \frac{1}{2}(L_+L_- + L_-L_+) = \frac{1}{4}H^2 + \frac{1}{2}(EF + FE).$$

5(a). Casimir eigenvalue on V_J

We compute the action of J^2 on $|J, M\rangle$. First,

$$L_z|J, M\rangle = M|J, M\rangle,$$

so

$$L_z^2|J, M\rangle = M^2|J, M\rangle.$$

Next,

$$L_-|J, M\rangle = \sqrt{(J+M)(J-M+1)}|J, M-1\rangle,$$

and applying L_+ afterwards gives

$$L_+L_-|J, M\rangle = (J+M)(J-M+1)|J, M\rangle.$$

Similarly,

$$L_- L_+ |J, M\rangle = (J - M)(J + M + 1) |J, M\rangle.$$

Therefore

$$\begin{aligned} J^2 |J, M\rangle &= \left[M^2 + \frac{1}{2}((J + M)(J - M + 1) + (J - M)(J + M + 1)) \right] |J, M\rangle \\ &= J(J + 1) |J, M\rangle. \end{aligned}$$

Thus

$$\boxed{J^2|_{V_J} = J(J + 1)\text{id.}}$$

For the orbital part V_1 , we have

$$\boxed{L^2 = 1(1 + 1)\text{id} = 2\text{id.}}$$

For the spin part $V_{1/2}$, we have

$$\boxed{S^2 = \frac{1}{2} \left(\frac{1}{2} + 1 \right) \text{id} = \frac{3}{4} \text{id.}}$$

5(b). Eigenvalues of $L \cdot S$

The spin-orbit operator is

$$L \cdot S = \frac{1}{2}((L + S)^2 - L^2 - S^2).$$

Using the decomposition

$$V_1 \otimes V_{1/2} \cong V_{3/2} \oplus V_{1/2},$$

the total Casimir $(L + S)^2$ acts as

$$J(J + 1)$$

on the summand V_J .

On $V_{3/2}$,

$$(L + S)^2 = \frac{3}{2} \left(\frac{3}{2} + 1 \right) = \frac{15}{4}.$$

Therefore

$$\begin{aligned} L \cdot S &= \frac{1}{2} \left(\frac{15}{4} - 2 - \frac{3}{4} \right) \\ &= \frac{1}{2} \left(\frac{15}{4} - \frac{11}{4} \right) \\ &= \frac{1}{2}. \end{aligned}$$

Thus

$$\boxed{L \cdot S = \frac{1}{2} \text{id} \quad \text{on } V_{3/2}.$$

On $V_{1/2}$,

$$(L + S)^2 = \frac{1}{2} \left(\frac{1}{2} + 1 \right) = \frac{3}{4}.$$

Therefore

$$\begin{aligned} L \cdot S &= \frac{1}{2} \left(\frac{3}{4} - 2 - \frac{3}{4} \right) \\ &= \frac{1}{2}(-2) \\ &= -1. \end{aligned}$$

Thus

$$\boxed{L \cdot S = -\text{id} \quad \text{on } V_{1/2}.}$$

5(c). Energy eigenvalues and degeneracies

The Hamiltonian is

$$H_{\text{SO}} = \lambda L \cdot S, \quad \lambda > 0.$$

Therefore the energy eigenvalues are:

summand	$L \cdot S$ eigenvalue	H_{SO} eigenvalue
$V_{3/2}$	$\frac{1}{2}$	$\frac{\lambda}{2}$
$V_{1/2}$	-1	$-\lambda$

The degeneracy of each energy level is the dimension of the corresponding irreducible representation:

$$\dim V_{3/2} = 4, \quad \dim V_{1/2} = 2.$$

Hence

$$\boxed{E_{3/2} = \frac{\lambda}{2} \text{ with degeneracy } 4,}$$

and

$$\boxed{E_{1/2} = -\lambda \text{ with degeneracy } 2.}$$

5(d). Good quantum numbers

The spin-orbit Hamiltonian depends on

$$(L + S)^2, \quad L^2, \quad S^2.$$

For fixed orbital angular momentum $\ell = 1$ and fixed spin $s = \frac{1}{2}$, L^2 and S^2 are already fixed scalars. Thus the nontrivial part is the total angular momentum operator $(L + S)^2$.

The coupled basis

$$\{|J, M\rangle\}$$

is built precisely to diagonalize

$$(L + S)^2 \quad \text{and} \quad L_z + S_z.$$

Therefore it diagonalizes $L \cdot S$, and hence also the spin-orbit Hamiltonian $H_{\text{SO}} = \lambda L \cdot S$.

By contrast, the tensor product basis

$$|1, m\rangle \otimes |\tfrac{1}{2}, m'\rangle$$

diagonalizes L_z and S_z separately, but not $(L + S)^2$. The reason is that

$$L \cdot S = L_z S_z + \frac{1}{2}(L_+ S_- + L_- S_+)$$

contains terms which change m and m' separately while keeping

$$M = m + m'$$

fixed. Therefore the tensor product basis is generally mixed by $L \cdot S$, whereas the coupled basis is diagonal.

5(e). Direct verification in the tensor product basis

We verify the two eigenvalues directly for the vectors

$$|\frac{3}{2}, \frac{1}{2}\rangle \quad \text{and} \quad |\frac{1}{2}, \frac{1}{2}\rangle.$$

Let

$$a := |1, 0\rangle \otimes |+\rangle, \quad b := |1, 1\rangle \otimes |-\rangle.$$

Then

$$|\frac{3}{2}, \frac{1}{2}\rangle = \frac{\sqrt{2}a + b}{\sqrt{3}},$$

and

$$|\frac{1}{2}, \frac{1}{2}\rangle = \frac{a - \sqrt{2}b}{\sqrt{3}}.$$

We use

$$L \cdot S = L_z S_z + \frac{1}{2}(L_+ S_- + L_- S_+).$$

First compute the action on a . Since $a = |1, 0\rangle \otimes |+\rangle$,

$$L_z S_z a = 0.$$

Also,

$$L_+ S_- a = \sqrt{2}b, \quad L_- S_+ a = 0.$$

Therefore

$$L \cdot S a = \frac{\sqrt{2}}{2}b.$$

Now compute the action on b . Since $b = |1, 1\rangle \otimes |-\rangle$,

$$L_z S_z b = 1 \cdot \left(-\frac{1}{2}\right)b = -\frac{1}{2}b.$$

Furthermore,

$$L_+ S_- b = 0, \quad L_- S_+ b = \sqrt{2}a.$$

Therefore

$$L \cdot S b = \frac{\sqrt{2}}{2}a - \frac{1}{2}b.$$

Now apply this to $|\frac{3}{2}, \frac{1}{2}\rangle$:

$$\begin{aligned}
L \cdot S |\frac{3}{2}, \frac{1}{2}\rangle &= L \cdot S \left(\frac{\sqrt{2}a + b}{\sqrt{3}} \right) \\
&= \frac{1}{\sqrt{3}} \left(\sqrt{2} \frac{\sqrt{2}}{2} b + \frac{\sqrt{2}}{2} a - \frac{1}{2} b \right) \\
&= \frac{1}{\sqrt{3}} \left(\frac{\sqrt{2}}{2} a + \frac{1}{2} b \right) \\
&= \frac{1}{2} \frac{\sqrt{2}a + b}{\sqrt{3}} \\
&= \frac{1}{2} |\frac{3}{2}, \frac{1}{2}\rangle.
\end{aligned}$$

So the eigenvalue is $\frac{1}{2}$, as predicted.

Finally, apply $L \cdot S$ to $|\frac{1}{2}, \frac{1}{2}\rangle$:

$$\begin{aligned}
L \cdot S |\frac{1}{2}, \frac{1}{2}\rangle &= L \cdot S \left(\frac{a - \sqrt{2}b}{\sqrt{3}} \right) \\
&= \frac{1}{\sqrt{3}} \left(\frac{\sqrt{2}}{2} b - \sqrt{2} \left(\frac{\sqrt{2}}{2} a - \frac{1}{2} b \right) \right) \\
&= \frac{1}{\sqrt{3}} (-a + \sqrt{2}b) \\
&= -\frac{a - \sqrt{2}b}{\sqrt{3}} \\
&= -|\frac{1}{2}, \frac{1}{2}\rangle.
\end{aligned}$$

So the eigenvalue is -1 , again agreeing with the Casimir computation.